General Relativity Analysis

PG-type warp metric (Spatial Conformal) Metric

Painleve-Gullstrand-type metric with lapse N=1, radial shift beta(r), and areal radius

SymPy GR Engine

February 25, 2026

Contents

Abstract

This document presents a complete symbolic General Relativity analysis of the **PG-type warp metric (Spatial Conformal)** metric. All quantities were computed analytically using SymPy. The report includes: the metric and its inverse, Christoffel symbols, Riemann and Ricci tensors, Ricci scalar, Einstein tensor, curvature invariants (Kretschmann scalar, Weyl tensor), orthonormal-frame stress-energy components, energy conditions, geodesic equations, and conservation law verification.

1 Metric Tensor and Conventions

1.1 Coordinate System and Conventions

Coordinates: $(t,\, r,\, \theta,\, \phi)$. Spacetime signature: $(-,+,+,+)$. Natural units: $G=c=1$. Greek indices run over $0,1,\ldots,3$; Latin hat-indices label orthonormal frame legs.

1.2 Covariant Metric $g_{\mu\nu}$

The line element is $ds^2 = g_{\mu\nu}\, dx^\mu dx^\nu$.

$$g_{\mu\nu} = \begin{pmatrix} B^2(r)\beta^2(r)-1 & B^2(r)\beta(r) & 0 & 0 \\ B^2(r)\beta(r) & B^2(r) & 0 & 0 \\ 0 & 0 & r^2B^2(r) & 0 \\ 0 & 0 & 0 & r^2B^2(r)\sin^2(\theta) \end{pmatrix} \tag{1}$$

1.3 Contravariant Metric $g^{\mu\nu}$

Defined by $g^{\mu\rho}\, g_{\rho\nu} = \delta^\mu{}_\nu$.

$$g^{\mu\nu} = \begin{pmatrix} -1 & \beta(r) & 0 & 0 \\ \beta(r) & \frac{-B^2(r)\beta^2(r)+1}{B^2(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2B^2(r)} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2B^2(r)\sin^2(\theta)} \end{pmatrix} \tag{2}$$

1.4 Metric Determinant

$$\det(g) = -r^4B^6(r)\sin^2(\theta)$$

2 Christoffel Symbols

The Christoffel symbols of the second kind (Levi-Civita connection) are

$$\Gamma^\lambda{}_{\mu\nu} = \frac{1}{2}\, g^{\lambda\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})\,. \tag{3}$$

They are symmetric in the lower pair: $\Gamma^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\nu\mu}$. Only non-zero independent components (with $\mu \le \nu$) are listed. Total non-zero independent components: **14**.

$$\Gamma^t{}_{tt} = -B^2(r)\beta^2(r)\frac{d}{dr}\beta(r) - B(r)\beta^3(r)\frac{d}{dr}B(r)$$

$$\Gamma^t{}_{tr} = -B^2(r)\beta(r)\frac{d}{dr}\beta(r) - B(r)\beta^2(r)\frac{d}{dr}B(r)$$

$$\Gamma^t{}_{rr} = -B^2(r)\frac{d}{dr}\beta(r) - B(r)\beta(r)\frac{d}{dr}B(r)$$

$$\Gamma^t{}_{\theta\theta} = -r^2B(r)\beta(r)\frac{d}{dr}B(r) - rB^2(r)\beta(r)$$

$$\Gamma^t{}_{\varphi\varphi} = -r^2B(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r) - rB^2(r)\beta(r)\sin^2(\theta)$$

$$\Gamma^{r}{}_{tt} = \frac{B^{3}(r)\beta^{3}(r)\frac{d}{dr}\beta(r) + B^{2}(r)\beta^{4}(r)\frac{d}{dr}B(r) - B(r)\beta(r)\frac{d}{dr}\beta(r) - \beta^{2}(r)\frac{d}{dr}B(r)}{B(r)}$$

$$\Gamma^{r}{}_{tr} = B^{2}(r)\beta^{2}(r)\frac{d}{dr}\beta(r) + B(r)\beta^{3}(r)\frac{d}{dr}B(r)$$

$$\Gamma^{r}{}_{rr} = \frac{B^{3}(r)\beta(r)\frac{d}{dr}\beta(r) + B^{2}(r)\beta^{2}(r)\frac{d}{dr}B(r) + \frac{d}{dr}B(r)}{B(r)}$$

$$\Gamma^{r}{}_{\theta\theta} = \frac{r^{2}B^{2}(r)\beta^{2}(r)\frac{d}{dr}B(r) - r^{2}\frac{d}{dr}B(r) + rB^{3}(r)\beta^{2}(r) - rB(r)}{B(r)}$$

$$\Gamma^{r}{}_{\varphi\varphi} = \frac{r^{2}B^{2}(r)\beta^{2}(r)\sin^{2}(\theta)\frac{d}{dr}B(r) - r^{2}\sin^{2}(\theta)\frac{d}{dr}B(r) + rB^{3}(r)\beta^{2}(r)\sin^{2}(\theta) - rB(r)\sin^{2}(\theta)}{B(r)}$$

$$\Gamma^{\theta}{}_{r\theta} = \frac{r\frac{d}{dr}B(r) + B(r)}{rB(r)}$$

$$\Gamma^{\theta}{}_{\varphi\varphi} = -\sin(\theta)\cos(\theta)$$

$$\Gamma^{\varphi}{}_{r\varphi} = \frac{r\frac{d}{dr}B(r) + B(r)}{rB(r)}$$

$$\Gamma^{\varphi}{}_{\theta\varphi} = \frac{\cos(\theta)}{\sin(\theta)}$$

3 Riemann Curvature Tensor

The Riemann tensor measures the failure of parallel transport around closed loops — the intrinsic curvature of spacetime:

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu}\Gamma^{\rho}{}_{\nu\sigma} - \partial_{\nu}\Gamma^{\rho}{}_{\mu\sigma} + \Gamma^{\rho}{}_{\mu\lambda}\Gamma^{\lambda}{}_{\nu\sigma} - \Gamma^{\rho}{}_{\nu\lambda}\Gamma^{\lambda}{}_{\mu\sigma}. \tag{4}$$

Antisymmetry: $R^{\rho}{}_{\sigma\mu\nu} = -R^{\rho}{}_{\sigma\nu\mu}$. Only components with $\mu < \nu$ are listed. Total independent non-zero Riemann components: **20**.

$$\begin{aligned} R^{t}{}_{ttr} = {} & B^{2}(r)\beta^{2}(r)\frac{d^{2}}{dr^{2}}\beta(r) \\ & + B^{2}(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^{2} \\ & + B(r)\beta^{3}(r)\frac{d^{2}}{dr^{2}}B(r) \\ & + 3B(r)\beta^{2}(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \end{aligned}$$

$$R^{t}{}_{rtr} = B^{2}(r)\beta(r)\frac{d^{2}}{dr^{2}}\beta(r) + B^{2}(r)\left(\frac{d}{dr}\beta(r)\right)^{2} + B(r)\beta^{2}(r)\frac{d^{2}}{dr^{2}}B(r) + 3B(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$

$$
\begin{aligned}
&{R^{t}}_{\theta t\theta}\\
&= r^{2} B(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) + r^{2}\beta^{2}(r)\left(\frac{d}{dr}B(r)\right)^{2} + rB^{2}(r)\beta(r)\frac{d}{dr}\beta(r) + rB(r)\beta^{2}(r)\frac{d}{dr}B(r)
\end{aligned}
$$

$$
{R^{t}}_{\theta r\theta} = -r^{2}B(r)\beta(r)\frac{d^{2}}{dr^{2}}B(r) + r^{2}\beta(r)\left(\frac{d}{dr}B(r)\right)^{2} - rB(r)\beta(r)\frac{d}{dr}B(r)
$$

$$
\begin{aligned}
{R^{t}}_{\varphi t\varphi} = {} & r^{2}B(r)\beta(r)\sin^{2}(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
& + r^{2}\beta^{2}(r)\sin^{2}(\theta)\left(\frac{d}{dr}B(r)\right)^{2}\\
& + rB^{2}(r)\beta(r)\sin^{2}(\theta)\frac{d}{dr}\beta(r)\\
& + rB(r)\beta^{2}(r)\sin^{2}(\theta)\frac{d}{dr}B(r)
\end{aligned}
$$

$$
{R^{t}}_{\varphi r\varphi} = -r^{2}B(r)\beta(r)\sin^{2}(\theta)\frac{d^{2}}{dr^{2}}B(r) + r^{2}\beta(r)\sin^{2}(\theta)\left(\frac{d}{dr}B(r)\right)^{2} - rB(r)\beta(r)\sin^{2}(\theta)\frac{d}{dr}B(r)
$$

$$
\begin{aligned}
{R^{r}}_{ttr} = {} & \frac{1}{B(r)}\Big(-B^{3}(r)\beta^{3}(r)\frac{d^{2}}{dr^{2}}\beta(r)\\
& - B^{3}(r)\beta^{2}(r)\left(\frac{d}{dr}\beta(r)\right)^{2}\\
& - B^{2}(r)\beta^{4}(r)\frac{d^{2}}{dr^{2}}B(r)\\
& - 3B^{2}(r)\beta^{3}(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
& + B(r)\beta(r)\frac{d^{2}}{dr^{2}}\beta(r)\\
& + B(r)\left(\frac{d}{dr}\beta(r)\right)^{2}\\
& + \beta^{2}(r)\frac{d^{2}}{dr^{2}}B(r)\\
& + 3\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
{R^{r}}_{rtr} = {} & -B^{2}(r)\beta^{2}(r)\frac{d^{2}}{dr^{2}}\beta(r)\\
& - B^{2}(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^{2}\\
& - B(r)\beta^{3}(r)\frac{d^{2}}{dr^{2}}B(r)\\
& - 3B(r)\beta^{2}(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)
\end{aligned}
$$

$$
\begin{aligned}
{R^{r}}_{\theta r\theta} = {} & \frac{1}{B^{2}(r)}\Big(r^{2}B^{3}(r)\beta^{2}(r)\frac{d^{2}}{dr^{2}}B(r)\\
& + r^{2}B^{3}(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)
\end{aligned}
$$

$$
\begin{aligned}
&- r^2 B(r) \frac{d^2}{dr^2} B(r) \\
&+ r^2 \left(\frac{d}{dr} B(r) \right)^2 \\
&+ r B^4(r) \beta(r) \frac{d}{dr} \beta(r) \\
&+ 2r B^3(r) \beta^2(r) \frac{d}{dr} B(r) \\
&- r B(r) \frac{d}{dr} B(r) \Big)
\end{aligned}
$$

$$
\begin{aligned}
R^r{}_{\varphi r \varphi} = \frac{1}{B^2(r)} \Big(& r^2 B^3(r) \beta^2(r) \sin^2(\theta) \frac{d^2}{dr^2} B(r) \\
&+ r^2 B^3(r) \beta(r) \sin^2(\theta) \frac{d}{dr} B(r) \frac{d}{dr} \beta(r) \\
&- r^2 B(r) \sin^2(\theta) \frac{d^2}{dr^2} B(r) \\
&+ r^2 \sin^2(\theta) \left(\frac{d}{dr} B(r) \right)^2 \\
&+ r B^4(r) \beta(r) \sin^2(\theta) \frac{d}{dr} \beta(r) \\
&+ 2r B^3(r) \beta^2(r) \sin^2(\theta) \frac{d}{dr} B(r) \\
&- r B(r) \sin^2(\theta) \frac{d}{dr} B(r) \Big)
\end{aligned}
$$

$$
\begin{aligned}
R^\theta{}_{t t \theta} = \frac{1}{r B^2(r)} \Big(& -r B^3(r) \beta^3(r) \frac{d}{dr} B(r) \frac{d}{dr} \beta(r) \\
&- r B^2(r) \beta^4(r) \left(\frac{d}{dr} B(r) \right)^2 \\
&+ r B(r) \beta(r) \frac{d}{dr} B(r) \frac{d}{dr} \beta(r) \\
&+ r \beta^2(r) \left(\frac{d}{dr} B(r) \right)^2 \\
&- B^4(r) \beta^3(r) \frac{d}{dr} \beta(r) \\
&- B^3(r) \beta^4(r) \frac{d}{dr} B(r) \\
&+ B^2(r) \beta(r) \frac{d}{dr} \beta(r) \\
&+ B(r) \beta^2(r) \frac{d}{dr} B(r) \Big)
\end{aligned}
$$

$$
R^\theta{}_{t r \theta} = \frac{-r B(r) \beta^2(r) \frac{d}{dr} B(r) \frac{d}{dr} \beta(r) - r \beta^3(r) \left(\frac{d}{dr} B(r) \right)^2 - B^2(r) \beta^2(r) \frac{d}{dr} \beta(r) - B(r) \beta^3(r) \frac{d}{dr} B(r)}{r}
$$

$$
R^\theta{}_{r t \theta} = \frac{-r B(r) \beta^2(r) \frac{d}{dr} B(r) \frac{d}{dr} \beta(r) - r \beta^3(r) \left(\frac{d}{dr} B(r) \right)^2 - B^2(r) \beta^2(r) \frac{d}{dr} \beta(r) - B(r) \beta^3(r) \frac{d}{dr} B(r)}{r}
$$

$$
\begin{aligned}
R^{\theta}{}_{rr\theta} = \frac{1}{rB^2(r)}\Big(&-rB^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&- rB^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2\\
&+ rB(r)\frac{d^2}{dr^2}B(r)\\
&- r\left(\frac{d}{dr}B(r)\right)^2\\
&- B^4(r)\beta(r)\frac{d}{dr}\beta(r)\\
&- B^3(r)\beta^2(r)\frac{d}{dr}B(r)\\
&+ B(r)\frac{d}{dr}B(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
R^{\theta}{}_{\varphi\theta\varphi} = \frac{1}{B^2(r)}\Big(&r^2B^2(r)\beta^2(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2\\
&- r^2\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2\\
&+ 2rB^3(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)\\
&- 2rB(r)\sin^2(\theta)\frac{d}{dr}B(r)\\
&+ B^4(r)\beta^2(r)\sin^2(\theta)\Big)
\end{aligned}
$$

$$
\begin{aligned}
R^{\varphi}{}_{tt\varphi} = \frac{1}{rB^2(r)}\Big(&-rB^3(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&- rB^2(r)\beta^4(r)\left(\frac{d}{dr}B(r)\right)^2\\
&+ rB(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&+ r\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2\\
&- B^4(r)\beta^3(r)\frac{d}{dr}\beta(r)\\
&- B^3(r)\beta^4(r)\frac{d}{dr}B(r)\\
&+ B^2(r)\beta(r)\frac{d}{dr}\beta(r)\\
&+ B(r)\beta^2(r)\frac{d}{dr}B(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
&R^{\varphi}{}_{tr\varphi}\\
&= \frac{-rB(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) - r\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2 - B^2(r)\beta^2(r)\frac{d}{dr}\beta(r) - B(r)\beta^3(r)\frac{d}{dr}B(r)}{r}
\end{aligned}
$$

$$
\begin{aligned}
&R^{\varphi}{}_{rt\varphi}\\
&= \frac{-rB(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) - r\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2 - B^2(r)\beta^2(r)\frac{d}{dr}\beta(r) - B(r)\beta^3(r)\frac{d}{dr}B(r)}{r}
\end{aligned}
$$

$$
\begin{aligned}
R^{\varphi}{}_{r r \varphi} = \frac{1}{rB^2(r)} \Big(& -rB^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
& - rB^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
& + rB(r)\frac{d^2}{dr^2}B(r) \\
& - r\left(\frac{d}{dr}B(r)\right)^2 \\
& - B^4(r)\beta(r)\frac{d}{dr}\beta(r) \\
& - B^3(r)\beta^2(r)\frac{d}{dr}B(r) \\
& + B(r)\frac{d}{dr}B(r)\Big)
\end{aligned}
$$

$$
R^{\varphi}{}_{\theta\theta\varphi} = \frac{-r^2B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 + r^2\left(\frac{d}{dr}B(r)\right)^2 - 2rB^3(r)\beta^2(r)\frac{d}{dr}B(r) + 2rB(r)\frac{d}{dr}B(r) - B^4(r)\beta^2(r)}{B^2(r)}
$$

4 Ricci Tensor

The Ricci tensor is the trace of the Riemann tensor: $R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$. It encodes how volumes change under parallel transport and is directly sourced by the stress-energy.

Non-zero components (matrix form exceeds page width):

$$
\begin{aligned}
R_{tt} = \frac{1}{rB^2(r)} \Big(& rB^4(r)\beta^3(r)\frac{d^2}{dr^2}\beta(r) \\
& + rB^4(r)\beta^2(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
& + rB^3(r)\beta^4(r)\frac{d^2}{dr^2}B(r) \\
& + 5rB^3(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
& + 2rB^2(r)\beta^4(r)\left(\frac{d}{dr}B(r)\right)^2 \\
& - rB^2(r)\beta(r)\frac{d^2}{dr^2}\beta(r) \\
& - rB^2(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
& - rB(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\
& - 5rB(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
& - 2r\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
& + 2B^4(r)\beta^3(r)\frac{d}{dr}\beta(r) \\
& + 2B^3(r)\beta^4(r)\frac{d}{dr}B(r)
\end{aligned}
$$

$$- 2B^2(r)\beta(r)\frac{d}{dr}\beta(r)$$
$$- 2B(r)\beta^2(r)\frac{d}{dr}B(r)\Big)$$

$$R_{tr} = \frac{1}{r}\Big(rB^2(r)\beta^2(r)\frac{d^2}{dr^2}\beta(r)$$
$$+ rB^2(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ rB(r)\beta^3(r)\frac{d^2}{dr^2}B(r)$$
$$+ 5rB(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$+ 2r\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$+ 2B^2(r)\beta^2(r)\frac{d}{dr}\beta(r)$$
$$+ 2B(r)\beta^3(r)\frac{d}{dr}B(r)\Big)$$

$$R_{rt} = \frac{1}{r}\Big(rB^2(r)\beta^2(r)\frac{d^2}{dr^2}\beta(r)$$
$$+ rB^2(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ rB(r)\beta^3(r)\frac{d^2}{dr^2}B(r)$$
$$+ 5rB(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$+ 2r\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$+ 2B^2(r)\beta^2(r)\frac{d}{dr}\beta(r)$$
$$+ 2B(r)\beta^3(r)\frac{d}{dr}B(r)\Big)$$

$$R_{rr} = \frac{1}{rB^2(r)}\Big(rB^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r)$$
$$+ rB^4(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ rB^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r)$$
$$+ 5rB^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$+ 2rB^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$- 2rB(r)\frac{d^2}{dr^2}B(r)$$
$$+ 2r\left(\frac{d}{dr}B(r)\right)^2$$

$$+ 2B^4(r)\beta(r)\frac{d}{dr}\beta(r)$$
$$+ 2B^3(r)\beta^2(r)\frac{d}{dr}B(r)$$
$$- 2B(r)\frac{d}{dr}B(r)\Big)$$

$$R_{\theta\theta} = \frac{1}{B(r)}\Big(r^2B^2(r)\beta^2(r)\frac{d^2}{dr^2}B(r)$$
$$+ 2r^2B^2(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$+ 2r^2B(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$- r^2\frac{d^2}{dr^2}B(r)$$
$$+ 2rB^3(r)\beta(r)\frac{d}{dr}\beta(r)$$
$$+ 5rB^2(r)\beta^2(r)\frac{d}{dr}B(r)$$
$$- 3r\frac{d}{dr}B(r)$$
$$+ B^3(r)\beta^2(r)\Big)$$

$$R_{\varphi\varphi} = \frac{1}{B(r)}\Big(r^2B^2(r)\beta^2(r)\sin^2(\theta)\frac{d^2}{dr^2}B(r)$$
$$+ 2r^2B^2(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$+ 2r^2B(r)\beta^2(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2$$
$$- r^2\sin^2(\theta)\frac{d^2}{dr^2}B(r)$$
$$+ 2rB^3(r)\beta(r)\sin^2(\theta)\frac{d}{dr}\beta(r)$$
$$+ 5rB^2(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)$$
$$- 3r\sin^2(\theta)\frac{d}{dr}B(r)$$
$$+ B^3(r)\beta^2(r)\sin^2(\theta)\Big)$$

5 Ricci Scalar (Curvature Scalar)

The Ricci scalar $\mathcal{R} = g^{\mu\nu}R_{\mu\nu}$ is the full contraction of the Ricci tensor. It provides a single-number measure of curvature. It vanishes for vacuum spacetimes.

Ricci scalar

$$\mathcal{R} = \frac{1}{r^2B^4(r)}\Big(2r^2B^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r)$$

$$
\begin{aligned}
&+ 2r^2 B^4(r) \left(\frac{d}{dr}\beta(r)\right)^2 \\
&+ 6r^2 B^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\
&+ 14r^2 B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
&+ 6r^2 B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
&- 4r^2 B(r)\frac{d^2}{dr^2}B(r) \\
&+ 2r^2\left(\frac{d}{dr}B(r)\right)^2 \\
&+ 8rB^4(r)\beta(r)\frac{d}{dr}\beta(r) \\
&+ 16rB^3(r)\beta^2(r)\frac{d}{dr}B(r) \\
&- 8rB(r)\frac{d}{dr}B(r) \\
&+ 2B^4(r)\beta^2(r)\Big)
\end{aligned}
$$

6 Einstein Tensor

The Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}\, g_{\mu\nu}\, \mathcal{R}$ is divergence-free by the contracted Bianchi identity. The Einstein field equations read $G_{\mu\nu} = 8\pi T_{\mu\nu}$.

Non-zero components (matrix form exceeds page width):

$$
\begin{aligned}
G_{tt} = \frac{1}{r^2 B^4(r)}\Big(&-2r^2 B^5(r)\beta^4(r)\frac{d^2}{dr^2}B(r) \\
&- 2r^2 B^5(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
&- r^2 B^4(r)\beta^4(r)\left(\frac{d}{dr}B(r)\right)^2 \\
&+ 4r^2 B^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\
&+ 2r^2 B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
&- 2r^2 B(r)\frac{d^2}{dr^2}B(r) \\
&+ r^2\left(\frac{d}{dr}B(r)\right)^2 \\
&- 2rB^6(r)\beta^3(r)\frac{d}{dr}\beta(r) \\
&- 6rB^5(r)\beta^4(r)\frac{d}{dr}B(r) \\
&+ 2rB^4(r)\beta(r)\frac{d}{dr}\beta(r) \\
&+ 10rB^3(r)\beta^2(r)\frac{d}{dr}B(r)
\end{aligned}
$$

$$
\begin{aligned}
&- 4rB(r)\frac{d}{dr}B(r)\\
&- B^6(r)\beta^4(r)\\
&+ B^4(r)\beta^2(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
G_{tr} = \frac{1}{r^2B^2(r)}\Big(&-2r^2B^3(r)\beta^3(r)\frac{d^2}{dr^2}B(r)\\
&- 2r^2B^3(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&- r^2B^2(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2\\
&+ 2r^2B(r)\beta(r)\frac{d^2}{dr^2}B(r)\\
&- r^2\beta(r)\left(\frac{d}{dr}B(r)\right)^2\\
&- 2rB^4(r)\beta^2(r)\frac{d}{dr}\beta(r)\\
&- 6rB^3(r)\beta^3(r)\frac{d}{dr}B(r)\\
&+ 4rB(r)\beta(r)\frac{d}{dr}B(r)\\
&- B^4(r)\beta^3(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
G_{rt} = \frac{1}{r^2B^2(r)}\Big(&-2r^2B^3(r)\beta^3(r)\frac{d^2}{dr^2}B(r)\\
&- 2r^2B^3(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&- r^2B^2(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2\\
&+ 2r^2B(r)\beta(r)\frac{d^2}{dr^2}B(r)\\
&- r^2\beta(r)\left(\frac{d}{dr}B(r)\right)^2\\
&- 2rB^4(r)\beta^2(r)\frac{d}{dr}\beta(r)\\
&- 6rB^3(r)\beta^3(r)\frac{d}{dr}B(r)\\
&+ 4rB(r)\beta(r)\frac{d}{dr}B(r)\\
&- B^4(r)\beta^3(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
G_{rr} = \frac{1}{r^2B^2(r)}\Big(&-2r^2B^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r)\\
&- 2r^2B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&- r^2B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2
\end{aligned}
$$

$$+ r^{2} \left(\frac{d}{dr} B(r)\right)^{2}$$

$$- 2 r B^{4}(r) \beta(r) \frac{d}{dr} \beta(r)$$

$$- 6 r B^{3}(r) \beta^{2}(r) \frac{d}{dr} B(r)$$

$$+ 2 r B(r) \frac{d}{dr} B(r)$$

$$- B^{4}(r) \beta^{2}(r)\Big)$$

$$G_{\theta\theta} = \frac{1}{B^{2}(r)} \Big(- r^{2} B^{4}(r) \beta(r) \frac{d^{2}}{dr^{2}} \beta(r)$$

$$- r^{2} B^{4}(r) \left(\frac{d}{dr} \beta(r)\right)^{2}$$

$$- 2 r^{2} B^{3}(r) \beta^{2}(r) \frac{d^{2}}{dr^{2}} B(r)$$

$$- 5 r^{2} B^{3}(r) \beta(r) \frac{d}{dr} B(r) \frac{d}{dr} \beta(r)$$

$$- r^{2} B^{2}(r) \beta^{2}(r) \left(\frac{d}{dr} B(r)\right)^{2}$$

$$+ r^{2} B(r) \frac{d^{2}}{dr^{2}} B(r)$$

$$- r^{2} \left(\frac{d}{dr} B(r)\right)^{2}$$

$$- 2 r B^{4}(r) \beta(r) \frac{d}{dr} \beta(r)$$

$$- 3 r B^{3}(r) \beta^{2}(r) \frac{d}{dr} B(r)$$

$$+ r B(r) \frac{d}{dr} B(r)\Big)$$

$$G_{\varphi\varphi} = \frac{1}{B^{2}(r)} \Big(- r^{2} B^{4}(r) \beta(r) \sin^{2}(\theta) \frac{d^{2}}{dr^{2}} \beta(r)$$

$$- r^{2} B^{4}(r) \sin^{2}(\theta) \left(\frac{d}{dr} \beta(r)\right)^{2}$$

$$- 2 r^{2} B^{3}(r) \beta^{2}(r) \sin^{2}(\theta) \frac{d^{2}}{dr^{2}} B(r)$$

$$- 5 r^{2} B^{3}(r) \beta(r) \sin^{2}(\theta) \frac{d}{dr} B(r) \frac{d}{dr} \beta(r)$$

$$- r^{2} B^{2}(r) \beta^{2}(r) \sin^{2}(\theta) \left(\frac{d}{dr} B(r)\right)^{2}$$

$$+ r^{2} B(r) \sin^{2}(\theta) \frac{d^{2}}{dr^{2}} B(r)$$

$$- r^{2} \sin^{2}(\theta) \left(\frac{d}{dr} B(r)\right)^{2}$$

$$- 2 r B^{4}(r) \beta(r) \sin^{2}(\theta) \frac{d}{dr} \beta(r)$$

$$- 3 r B^{3}(r) \beta^{2}(r) \sin^{2}(\theta) \frac{d}{dr} B(r)$$

$$+ rB(r)\sin^2(\theta)\frac{d}{dr}B(r)\Big)$$

7 Curvature Invariants

7.1 Kretschmann Scalar

The Kretschmann scalar $\mathcal{K} = R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}$ is a quadratic curvature invariant. Unlike $\mathcal{R}$, it remains nonzero even in vacuum. It diverges at genuine (physical) curvature singularities, distinguishing them from coordinate singularities. For the Schwarzschild metric: $\mathcal{K} = 48M^2/r^6$.

Kretschmann scalar

$$\begin{aligned}
\mathcal{K} = \frac{1}{r^4B^8(r)}\Big(&4r^4B^8(r)\beta^2(r)\left(\frac{d^2}{dr^2}\beta(r)\right)^2 \\
&+ 8r^4B^8(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^2\frac{d^2}{dr^2}\beta(r) \\
&+ 4r^4B^8(r)\left(\frac{d}{dr}\beta(r)\right)^4 \\
&+ 8r^4B^7(r)\beta^3(r)\frac{d^2}{dr^2}B(r)\frac{d^2}{dr^2}\beta(r) \\
&+ 24r^4B^7(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\frac{d^2}{dr^2}\beta(r) \\
&+ 8r^4B^7(r)\beta^2(r)\frac{d^2}{dr^2}B(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
&+ 24r^4B^7(r)\beta(r)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^3 \\
&+ 12r^4B^6(r)\beta^4(r)\left(\frac{d^2}{dr^2}B(r)\right)^2 \\
&+ 40r^4B^6(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)\frac{d}{dr}\beta(r) \\
&+ 52r^4B^6(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2\left(\frac{d}{dr}\beta(r)\right)^2 \\
&+ 16r^4B^5(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^3\frac{d}{dr}\beta(r) \\
&+ 12r^4B^4(r)\beta^4(r)\left(\frac{d}{dr}B(r)\right)^4 \\
&- 16r^4B^4(r)\beta^2(r)\left(\frac{d^2}{dr^2}B(r)\right)^2 \\
&- 16r^4B^4(r)\beta(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)\frac{d}{dr}\beta(r) \\
&+ 16r^4B^3(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d^2}{dr^2}B(r) \\
&+ 16r^4B^3(r)\beta(r)\left(\frac{d}{dr}B(r)\right)^3\frac{d}{dr}\beta(r) \\
&- 8r^4B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^4
\end{aligned}$$

$$
\begin{aligned}
&+ 8r^4 B^2(r) \left(\frac{d^2}{dr^2} B(r)\right)^2 \\
&- 16r^4 B(r) \left(\frac{d}{dr} B(r)\right)^2 \frac{d^2}{dr^2} B(r) \\
&+ 12r^4 \left(\frac{d}{dr} B(r)\right)^4 \\
&+ 16r^3 B^7(r) \beta^3(r) \frac{d^2}{dr^2} B(r) \frac{d}{dr} \beta(r) \\
&+ 32r^3 B^7(r) \beta^2(r) \frac{d}{dr} B(r) \left(\frac{d}{dr} \beta(r)\right)^2 \\
&+ 32r^3 B^6(r) \beta^4(r) \frac{d}{dr} B(r) \frac{d^2}{dr^2} B(r) \\
&+ 64r^3 B^6(r) \beta^3(r) \left(\frac{d}{dr} B(r)\right)^2 \frac{d}{dr} \beta(r) \\
&+ 32r^3 B^5(r) \beta^4(r) \left(\frac{d}{dr} B(r)\right)^3 \\
&- 16r^3 B^5(r) \beta(r) \frac{d^2}{dr^2} B(r) \frac{d}{dr} \beta(r) \\
&- 48r^3 B^4(r) \beta^2(r) \frac{d}{dr} B(r) \frac{d^2}{dr^2} B(r) \\
&+ 16r^3 B^2(r) \frac{d}{dr} B(r) \frac{d^2}{dr^2} B(r) \\
&+ 16r^2 B^8(r) \beta^2(r) \left(\frac{d}{dr} \beta(r)\right)^2 \\
&+ 48r^2 B^7(r) \beta^3(r) \frac{d}{dr} B(r) \frac{d}{dr} \beta(r) \\
&+ 64r^2 B^6(r) \beta^4(r) \left(\frac{d}{dr} B(r)\right)^2 \\
&- 16r^2 B^5(r) \beta(r) \frac{d}{dr} B(r) \frac{d}{dr} \beta(r) \\
&- 72r^2 B^4(r) \beta^2(r) \left(\frac{d}{dr} B(r)\right)^2 \\
&+ 24r^2 B^2(r) \left(\frac{d}{dr} B(r)\right)^2 \\
&+ 16r B^7(r) \beta^4(r) \frac{d}{dr} B(r) \\
&- 16r B^5(r) \beta^2(r) \frac{d}{dr} B(r) \\
&+ 4B^8(r) \beta^4(r) \Big)
\end{aligned}
$$

7.2 Weyl Conformal Tensor

The Weyl tensor $C_{\rho\sigma\mu\nu}$ is the trace-free part of the Riemann tensor, representing pure gravitational (tidal / wave) degrees of freedom independent of the local matter distribution. It vanishes in conformally flat spacetimes and is equal to the Riemann tensor in vacuum. Independent non-zero

Weyl components: **20**.

$$
\begin{aligned}
C_{t\,r\,t\,r} = \frac{1}{3r^2B^2(r)}\Big(&-r^2B^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r) \\
&- r^2B^4(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
&- r^2B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
&- r^2B(r)\frac{d^2}{dr^2}B(r) \\
&+ 2r^2\left(\frac{d}{dr}B(r)\right)^2 \\
&+ 2rB^4(r)\beta(r)\frac{d}{dr}\beta(r) \\
&+ rB^3(r)\beta^2(r)\frac{d}{dr}B(r) \\
&+ rB(r)\frac{d}{dr}B(r) \\
&- B^4(r)\beta^2(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
C_{t\,\theta\,t\,\theta} = \frac{1}{6B^2(r)}\Big(&-r^2B^6(r)\beta^3(r)\frac{d^2}{dr^2}\beta(r) \\
&- r^2B^6(r)\beta^2(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
&- r^2B^5(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
&+ r^2B^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r) \\
&+ r^2B^4(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
&- r^2B^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\
&+ r^2B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
&+ 2r^2B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
&+ r^2B(r)\frac{d^2}{dr^2}B(r) \\
&- 2r^2\left(\frac{d}{dr}B(r)\right)^2 \\
&+ 2rB^6(r)\beta^3(r)\frac{d}{dr}\beta(r) \\
&+ rB^5(r)\beta^4(r)\frac{d}{dr}B(r) \\
&- 2rB^4(r)\beta(r)\frac{d}{dr}\beta(r) \\
&- rB(r)\frac{d}{dr}B(r) \\
&- B^6(r)\beta^4(r)
\end{aligned}
$$

$$+ B^{4}(r)\beta^{2}(r)\Big)$$

$$C_{t\,\theta\,r\,\theta} = -\frac{r^{2}B^{4}(r)\beta^{2}(r)\frac{d^{2}}{dr^{2}}\beta(r)}{6} - \frac{r^{2}B^{4}(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^{2}}{6} - \frac{r^{2}B^{3}(r)\beta^{2}(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{6} - \frac{r^{2}B(r)\beta(r)\frac{d^{2}}{dr^{2}}B(r)}{6} + \frac{r^{2}\beta(r)\left(\frac{d}{dr}B(r)\right)^{2}}{3} + \frac{rB^{4}(r)\beta^{2}(r)\frac{d}{dr}\beta(r)}{3} + \frac{rB^{3}(r)\beta^{3}(r)\frac{d}{dr}B(r)}{6} + \frac{rB(r)\beta(r)\frac{d}{dr}B(r)}{6} - \frac{B^{4}(r)\beta^{3}(r)}{6}$$

$$C_{t\,\varphi\,t\,\varphi} = \frac{1}{6B^{2}(r)}\Big(-r^{2}B^{6}(r)\beta^{3}(r)\sin^{2}(\theta)\frac{d^{2}}{dr^{2}}\beta(r) - r^{2}B^{6}(r)\beta^{2}(r)\sin^{2}(\theta)\left(\frac{d}{dr}\beta(r)\right)^{2} - r^{2}B^{5}(r)\beta^{3}(r)\sin^{2}(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) + r^{2}B^{4}(r)\beta(r)\sin^{2}(\theta)\frac{d^{2}}{dr^{2}}\beta(r) + r^{2}B^{4}(r)\sin^{2}(\theta)\left(\frac{d}{dr}\beta(r)\right)^{2} - r^{2}B^{3}(r)\beta^{2}(r)\sin^{2}(\theta)\frac{d^{2}}{dr^{2}}B(r) + r^{2}B^{3}(r)\beta(r)\sin^{2}(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) + 2r^{2}B^{2}(r)\beta^{2}(r)\sin^{2}(\theta)\left(\frac{d}{dr}B(r)\right)^{2} + r^{2}B(r)\sin^{2}(\theta)\frac{d^{2}}{dr^{2}}B(r) - 2r^{2}\sin^{2}(\theta)\left(\frac{d}{dr}B(r)\right)^{2} + 2rB^{6}(r)\beta^{3}(r)\sin^{2}(\theta)\frac{d}{dr}\beta(r) + rB^{5}(r)\beta^{4}(r)\sin^{2}(\theta)\frac{d}{dr}B(r) - 2rB^{4}(r)\beta(r)\sin^{2}(\theta)\frac{d}{dr}\beta(r)$$

$$
\begin{aligned}
&- rB(r)\sin^2(\theta)\frac{d}{dr}B(r)\\
&- B^6(r)\beta^4(r)\sin^2(\theta)\\
&+ B^4(r)\beta^2(r)\sin^2(\theta)\Big)
\end{aligned}
$$

$$
\begin{aligned}
C_{t\varphi r\varphi} = &-\frac{r^2B^4(r)\beta^2(r)\sin^2(\theta)\frac{d^2}{dr^2}\beta(r)}{6}\\
&-\frac{r^2B^4(r)\beta(r)\sin^2(\theta)\left(\frac{d}{dr}\beta(r)\right)^2}{6}\\
&-\frac{r^2B^3(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{6}\\
&-\frac{r^2B(r)\beta(r)\sin^2(\theta)\frac{d^2}{dr^2}B(r)}{6}\\
&+\frac{r^2\beta(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2}{3}\\
&+\frac{rB^4(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}\beta(r)}{3}\\
&+\frac{rB^3(r)\beta^3(r)\sin^2(\theta)\frac{d}{dr}B(r)}{6}\\
&+\frac{rB(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)}{6}\\
&-\frac{B^4(r)\beta^3(r)\sin^2(\theta)}{6}
\end{aligned}
$$

$$
\begin{aligned}
C_{rttr} = \frac{1}{3r^2B^2(r)}\Big(&r^2B^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r)\\
&+ r^2B^4(r)\left(\frac{d}{dr}\beta(r)\right)^2\\
&+ r^2B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&+ r^2B(r)\frac{d^2}{dr^2}B(r)\\
&- 2r^2\left(\frac{d}{dr}B(r)\right)^2\\
&- 2rB^4(r)\beta(r)\frac{d}{dr}\beta(r)\\
&- rB^3(r)\beta^2(r)\frac{d}{dr}B(r)\\
&- rB(r)\frac{d}{dr}B(r)\\
&+ B^4(r)\beta^2(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
C_{r\theta t\theta} = &-\frac{r^2B^4(r)\beta^2(r)\frac{d^2}{dr^2}\beta(r)}{6}\\
&-\frac{r^2B^4(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^2}{6}
\end{aligned}
$$

$$- \frac{r^2 B^3(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{6}$$
$$- \frac{r^2 B(r)\beta(r)\frac{d^2}{dr^2}B(r)}{6}$$
$$+ \frac{r^2 \beta(r)\left(\frac{d}{dr}B(r)\right)^2}{3}$$
$$+ \frac{rB^4(r)\beta^2(r)\frac{d}{dr}\beta(r)}{3}$$
$$+ \frac{rB^3(r)\beta^3(r)\frac{d}{dr}B(r)}{6}$$
$$+ \frac{rB(r)\beta(r)\frac{d}{dr}B(r)}{6}$$
$$- \frac{B^4(r)\beta^3(r)}{6}$$

$$C_{r\,\theta\,r\,\theta} = -\frac{r^2 B^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r)}{6}$$
$$- \frac{r^2 B^4(r)\left(\frac{d}{dr}\beta(r)\right)^2}{6}$$
$$- \frac{r^2 B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{6}$$
$$- \frac{r^2 B(r)\frac{d^2}{dr^2}B(r)}{6}$$
$$+ \frac{r^2\left(\frac{d}{dr}B(r)\right)^2}{3}$$
$$+ \frac{rB^4(r)\beta(r)\frac{d}{dr}\beta(r)}{3}$$
$$+ \frac{rB^3(r)\beta^2(r)\frac{d}{dr}B(r)}{6}$$
$$+ \frac{rB(r)\frac{d}{dr}B(r)}{6}$$
$$- \frac{B^4(r)\beta^2(r)}{6}$$

$$C_{r\,\varphi\,t\,\varphi} = -\frac{r^2 B^4(r)\beta^2(r)\sin^2(\theta)\frac{d^2}{dr^2}\beta(r)}{6}$$
$$- \frac{r^2 B^4(r)\beta(r)\sin^2(\theta)\left(\frac{d}{dr}\beta(r)\right)^2}{6}$$
$$- \frac{r^2 B^3(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{6}$$
$$- \frac{r^2 B(r)\beta(r)\sin^2(\theta)\frac{d^2}{dr^2}B(r)}{6}$$
$$+ \frac{r^2\beta(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2}{3}$$
$$+ \frac{rB^4(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}\beta(r)}{3}$$

$$+\frac{rB^3(r)\beta^3(r)\sin^2(\theta)\frac{d}{dr}B(r)}{6}$$
$$+\frac{rB(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)}{6}$$
$$-\frac{B^4(r)\beta^3(r)\sin^2(\theta)}{6}$$

$$C_{r\varphi r\varphi} = -\frac{r^2B^4(r)\beta(r)\sin^2(\theta)\frac{d^2}{dr^2}\beta(r)}{6}$$
$$-\frac{r^2B^4(r)\sin^2(\theta)\left(\frac{d}{dr}\beta(r)\right)^2}{6}$$
$$-\frac{r^2B^3(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{6}$$
$$-\frac{r^2B(r)\sin^2(\theta)\frac{d^2}{dr^2}B(r)}{6}$$
$$+\frac{r^2\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2}{3}$$
$$+\frac{rB^4(r)\beta(r)\sin^2(\theta)\frac{d}{dr}\beta(r)}{3}$$
$$+\frac{rB^3(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)}{6}$$
$$+\frac{rB(r)\sin^2(\theta)\frac{d}{dr}B(r)}{6}$$
$$-\frac{B^4(r)\beta^2(r)\sin^2(\theta)}{6}$$

$$C_{\theta t t\theta} = \frac{1}{6B^2(r)}\Big(r^2B^6(r)\beta^3(r)\frac{d^2}{dr^2}\beta(r)$$
$$+r^2B^6(r)\beta^2(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+r^2B^5(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$-r^2B^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r)$$
$$-r^2B^4(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+r^2B^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r)$$
$$-r^2B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$-2r^2B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$-r^2B(r)\frac{d^2}{dr^2}B(r)$$
$$+2r^2\left(\frac{d}{dr}B(r)\right)^2$$
$$-2rB^6(r)\beta^3(r)\frac{d}{dr}\beta(r)$$

$$
\begin{aligned}
&- rB^5(r)\beta^4(r)\frac{d}{dr}B(r) \\
&+ 2rB^4(r)\beta(r)\frac{d}{dr}\beta(r) \\
&+ rB(r)\frac{d}{dr}B(r) \\
&+ B^6(r)\beta^4(r) \\
&- B^4(r)\beta^2(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
C_{\theta\, t\, r\, \theta} = & \frac{r^2B^4(r)\beta^2(r)\frac{d^2}{dr^2}\beta(r)}{6} \\
& + \frac{r^2B^4(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^2}{6} \\
& + \frac{r^2B^3(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{6} \\
& + \frac{r^2B(r)\beta(r)\frac{d^2}{dr^2}B(r)}{6} \\
& - \frac{r^2\beta(r)\left(\frac{d}{dr}B(r)\right)^2}{3} \\
& - \frac{rB^4(r)\beta^2(r)\frac{d}{dr}\beta(r)}{3} \\
& - \frac{rB^3(r)\beta^3(r)\frac{d}{dr}B(r)}{6} \\
& - \frac{rB(r)\beta(r)\frac{d}{dr}B(r)}{6} \\
& + \frac{B^4(r)\beta^3(r)}{6}
\end{aligned}
$$

$$
\begin{aligned}
C_{\theta\, r\, t\, \theta} = & \frac{r^2B^4(r)\beta^2(r)\frac{d^2}{dr^2}\beta(r)}{6} \\
& + \frac{r^2B^4(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^2}{6} \\
& + \frac{r^2B^3(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{6} \\
& + \frac{r^2B(r)\beta(r)\frac{d^2}{dr^2}B(r)}{6} \\
& - \frac{r^2\beta(r)\left(\frac{d}{dr}B(r)\right)^2}{3} \\
& - \frac{rB^4(r)\beta^2(r)\frac{d}{dr}\beta(r)}{3} \\
& - \frac{rB^3(r)\beta^3(r)\frac{d}{dr}B(r)}{6} \\
& - \frac{rB(r)\beta(r)\frac{d}{dr}B(r)}{6} \\
& + \frac{B^4(r)\beta^3(r)}{6}
\end{aligned}
$$

$$C_{\theta r r \theta} = \frac{r^2 B^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r)}{6}$$
$$+ \frac{r^2 B^4(r)\left(\frac{d}{dr}\beta(r)\right)^2}{6}$$
$$+ \frac{r^2 B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{6}$$
$$+ \frac{r^2 B(r)\frac{d^2}{dr^2}B(r)}{6}$$
$$- \frac{r^2\left(\frac{d}{dr}B(r)\right)^2}{3}$$
$$- \frac{rB^4(r)\beta(r)\frac{d}{dr}\beta(r)}{3}$$
$$- \frac{rB^3(r)\beta^2(r)\frac{d}{dr}B(r)}{6}$$
$$- \frac{rB(r)\frac{d}{dr}B(r)}{6}$$
$$+ \frac{B^4(r)\beta^2(r)}{6}$$

$$C_{\theta \varphi \theta \varphi} = \frac{r^4 B^4(r)\beta(r)\sin^2(\theta)\frac{d^2}{dr^2}\beta(r)}{3}$$
$$+ \frac{r^4 B^4(r)\sin^2(\theta)\left(\frac{d}{dr}\beta(r)\right)^2}{3}$$
$$+ \frac{r^4 B^3(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{3}$$
$$+ \frac{r^4 B(r)\sin^2(\theta)\frac{d^2}{dr^2}B(r)}{3}$$
$$- \frac{2r^4\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2}{3}$$
$$- \frac{2r^3 B^4(r)\beta(r)\sin^2(\theta)\frac{d}{dr}\beta(r)}{3}$$
$$- \frac{r^3 B^3(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)}{3}$$
$$- \frac{r^3 B(r)\sin^2(\theta)\frac{d}{dr}B(r)}{3}$$
$$+ \frac{r^2 B^4(r)\beta^2(r)\sin^2(\theta)}{3}$$

$$C_{\varphi t t \varphi} = \frac{1}{6B^2(r)}\Big(r^2 B^6(r)\beta^3(r)\sin^2(\theta)\frac{d^2}{dr^2}\beta(r)$$
$$+ r^2 B^6(r)\beta^2(r)\sin^2(\theta)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ r^2 B^5(r)\beta^3(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$- r^2 B^4(r)\beta(r)\sin^2(\theta)\frac{d^2}{dr^2}\beta(r)$$

$$
\begin{aligned}
&- r^2 B^4(r) \sin^2(\theta) \left(\frac{d}{dr}\beta(r)\right)^2 \\
&+ r^2 B^3(r)\beta^2(r) \sin^2(\theta) \frac{d^2}{dr^2} B(r) \\
&- r^2 B^3(r)\beta(r) \sin^2(\theta) \frac{d}{dr} B(r) \frac{d}{dr}\beta(r) \\
&- 2r^2 B^2(r)\beta^2(r) \sin^2(\theta) \left(\frac{d}{dr} B(r)\right)^2 \\
&- r^2 B(r) \sin^2(\theta) \frac{d^2}{dr^2} B(r) \\
&+ 2r^2 \sin^2(\theta) \left(\frac{d}{dr} B(r)\right)^2 \\
&- 2r B^6(r)\beta^3(r) \sin^2(\theta) \frac{d}{dr}\beta(r) \\
&- r B^5(r)\beta^4(r) \sin^2(\theta) \frac{d}{dr} B(r) \\
&+ 2r B^4(r)\beta(r) \sin^2(\theta) \frac{d}{dr}\beta(r) \\
&+ r B(r) \sin^2(\theta) \frac{d}{dr} B(r) \\
&+ B^6(r)\beta^4(r) \sin^2(\theta) \\
&- B^4(r)\beta^2(r) \sin^2(\theta) \Big)
\end{aligned}
$$

$$
\begin{aligned}
C_{\varphi t r \varphi} = & \frac{r^2 B^4(r)\beta^2(r) \sin^2(\theta) \frac{d^2}{dr^2}\beta(r)}{6} \\
&+ \frac{r^2 B^4(r)\beta(r) \sin^2(\theta) \left(\frac{d}{dr}\beta(r)\right)^2}{6} \\
&+ \frac{r^2 B^3(r)\beta^2(r) \sin^2(\theta) \frac{d}{dr} B(r) \frac{d}{dr}\beta(r)}{6} \\
&+ \frac{r^2 B(r)\beta(r) \sin^2(\theta) \frac{d^2}{dr^2} B(r)}{6} \\
&- \frac{r^2 \beta(r) \sin^2(\theta) \left(\frac{d}{dr} B(r)\right)^2}{3} \\
&- \frac{r B^4(r)\beta^2(r) \sin^2(\theta) \frac{d}{dr}\beta(r)}{3} \\
&- \frac{r B^3(r)\beta^3(r) \sin^2(\theta) \frac{d}{dr} B(r)}{6} \\
&- \frac{r B(r)\beta(r) \sin^2(\theta) \frac{d}{dr} B(r)}{6} \\
&+ \frac{B^4(r)\beta^3(r) \sin^2(\theta)}{6}
\end{aligned}
$$

$$
\begin{aligned}
C_{\varphi r t \varphi} = & \frac{r^2 B^4(r)\beta^2(r) \sin^2(\theta) \frac{d^2}{dr^2}\beta(r)}{6} \\
&+ \frac{r^2 B^4(r)\beta(r) \sin^2(\theta) \left(\frac{d}{dr}\beta(r)\right)^2}{6} \\
&+ \frac{r^2 B^3(r)\beta^2(r) \sin^2(\theta) \frac{d}{dr} B(r) \frac{d}{dr}\beta(r)}{6}
\end{aligned}
$$

$$+\frac{r^2B(r)\beta(r)\sin^2(\theta)\frac{d^2}{dr^2}B(r)}{6}$$
$$-\frac{r^2\beta(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2}{3}$$
$$-\frac{rB^4(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}\beta(r)}{3}$$
$$-\frac{rB^3(r)\beta^3(r)\sin^2(\theta)\frac{d}{dr}B(r)}{6}$$
$$-\frac{rB(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)}{6}$$
$$+\frac{B^4(r)\beta^3(r)\sin^2(\theta)}{6}$$

$$C_{\varphi r r \varphi}=\frac{r^2B^4(r)\beta(r)\sin^2(\theta)\frac{d^2}{dr^2}\beta(r)}{6}$$
$$+\frac{r^2B^4(r)\sin^2(\theta)\left(\frac{d}{dr}\beta(r)\right)^2}{6}$$
$$+\frac{r^2B^3(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{6}$$
$$+\frac{r^2B(r)\sin^2(\theta)\frac{d^2}{dr^2}B(r)}{6}$$
$$-\frac{r^2\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2}{3}$$
$$-\frac{rB^4(r)\beta(r)\sin^2(\theta)\frac{d}{dr}\beta(r)}{3}$$
$$-\frac{rB^3(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)}{6}$$
$$-\frac{rB(r)\sin^2(\theta)\frac{d}{dr}B(r)}{6}$$
$$+\frac{B^4(r)\beta^2(r)\sin^2(\theta)}{6}$$

$$C_{\varphi\theta\theta\varphi}=-\frac{r^4B^4(r)\beta(r)\sin^2(\theta)\frac{d^2}{dr^2}\beta(r)}{3}$$
$$-\frac{r^4B^4(r)\sin^2(\theta)\left(\frac{d}{dr}\beta(r)\right)^2}{3}$$
$$-\frac{r^4B^3(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)}{3}$$
$$-\frac{r^4B(r)\sin^2(\theta)\frac{d^2}{dr^2}B(r)}{3}$$
$$+\frac{2r^4\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2}{3}$$
$$+\frac{2r^3B^4(r)\beta(r)\sin^2(\theta)\frac{d}{dr}\beta(r)}{3}$$
$$+\frac{r^3B^3(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)}{3}$$

$$+\frac{r^{3} B(r) \sin^{2}(\theta) \frac{d}{dr} B(r)}{3}$$
$$-\frac{r^{2} B^{4}(r) \beta^{2}(r) \sin^{2}(\theta)}{3}$$

8 Orthonormal Frame Analysis

8.1 Tetrad Construction

The metric has an **ADM shift vector** with a diagonal spatial block ($\gamma_{ij} = 0$ for $i \neq j$). The tetrad was built via the ADM decomposition: lapse N, shift β^i, and spatial triad $E^{\hat{a}}{}_i = \sqrt{\gamma_{ii}}\,\delta^{\hat{a}}_i$ (no sum).

8.2 ADM Lapse and Shift

The ADM decomposition gives lapse N and shift β^i:

$$N = 1$$

$$\beta^r = \beta(r)$$

$$\beta^\theta = 0$$

$$\beta^\varphi = 0$$

8.3 Spatial Triad $E^{\hat{a}}{}_i$

The spatial triad satisfies $\gamma_{ij} = \delta_{\hat{a}\hat{b}}\, E^{\hat{a}}{}_i\, E^{\hat{b}}{}_j$.

$$E^{\hat{a}}{}_i = \begin{pmatrix} B(r) & 0 & 0 \\ 0 & B(r)\,|r| & 0 \\ 0 & 0 & B(r)\,|r|\,|\sin(\theta)| \end{pmatrix} \tag{5}$$

8.4 Contravariant Tetrad $e^{\mu}{}_{\hat{a}}$

Entry $[\mu,\, \hat{a}]$ of the matrix below is $e^{\mu}{}_{\hat{a}}$. It satisfies $g_{\mu\nu}\, e^{\mu}{}_{\hat{a}}\, e^{\nu}{}_{\hat{b}} = \eta_{\hat{a}\hat{b}}$.

$$e^{\mu}{}_{\hat{a}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -\beta(r) & \frac{1}{B(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{B(r)|r|} & 0 \\ 0 & 0 & 0 & \frac{1}{B(r)|r||\sin(\theta)|} \end{pmatrix} \tag{6}$$

8.5 Coframe $e^{\hat{a}}{}_{\mu}$

Entry $[\hat{a},\, \mu]$ of the matrix below is $e^{\hat{a}}{}_{\mu}$ (the dual frame / coframe).

$$e^{\hat{a}}{}_{\mu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ B(r)\beta(r) & B(r) & 0 & 0 \\ 0 & 0 & B(r)\,|r| & 0 \\ 0 & 0 & 0 & B(r)\,|r|\,|\sin(\theta)| \end{pmatrix} \tag{7}$$

8.6 Orthonormality Verification

Orthonormality Verified

$$g_{\mu\nu}\, e^{\mu}{}_{\hat{a}}\, e^{\nu}{}_{\hat{b}} = \eta_{\hat{a}\hat{b}} = \mathrm{diag}(-1, +1, +1, +1) \qquad \checkmark$$

8.7 Orthonormal-Frame Einstein Tensor $\hat{G}_{\hat{a}\hat{b}}$

The orthonormal-frame Einstein tensor is defined by $\hat{G}_{\hat{a}\hat{b}} = e^{\mu}{}_{\hat{a}}\, e^{\nu}{}_{\hat{b}}\, G_{\mu\nu}$.
Non-zero components (matrix form exceeds page width):

$$\begin{aligned}\hat{G}_{\hat{t}\hat{t}} = \frac{1}{r^2 B^4(r)} \Big(& 2r^2 B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\ & + 3r^2 B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\ & - 2r^2 B(r)\frac{d^2}{dr^2}B(r) \\ & + r^2\left(\frac{d}{dr}B(r)\right)^2 \\ & + 2rB^4(r)\beta(r)\frac{d}{dr}\beta(r) \\ & + 4rB^3(r)\beta^2(r)\frac{d}{dr}B(r) \\ & - 4rB(r)\frac{d}{dr}B(r) \\ & + B^4(r)\beta^2(r)\Big)\end{aligned}$$

$$\hat{G}_{\hat{t}\hat{r}} = \frac{2rB(r)\beta(r)\frac{d^2}{dr^2}B(r) - 2r\beta(r)\left(\frac{d}{dr}B(r)\right)^2 + 2B(r)\beta(r)\frac{d}{dr}B(r)}{rB^3(r)}$$

$$\hat{G}_{\hat{r}\hat{t}} = \frac{2rB(r)\beta(r)\frac{d^2}{dr^2}B(r) - 2r\beta(r)\left(\frac{d}{dr}B(r)\right)^2 + 2B(r)\beta(r)\frac{d}{dr}B(r)}{rB^3(r)}$$

$$\begin{aligned}\hat{G}_{\hat{r}\hat{r}} = \frac{1}{r^2 B^4(r)} \Big(& -2r^2 B^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\ & - 2r^2 B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\ & - r^2 B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\ & + r^2\left(\frac{d}{dr}B(r)\right)^2 \\ & - 2rB^4(r)\beta(r)\frac{d}{dr}\beta(r) \\ & - 6rB^3(r)\beta^2(r)\frac{d}{dr}B(r) \\ & + 2rB(r)\frac{d}{dr}B(r) \\ & - B^4(r)\beta^2(r)\Big)\end{aligned}$$

$$
\begin{aligned}
\hat{G}_{\hat{\theta}\hat{\theta}} = \frac{1}{rB^4(r)}\Big(&-rB^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r) \\
&- rB^4(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
&- 2rB^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\
&- 5rB^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
&- rB^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
&+ rB(r)\frac{d^2}{dr^2}B(r) \\
&- r\left(\frac{d}{dr}B(r)\right)^2 \\
&- 2B^4(r)\beta(r)\frac{d}{dr}\beta(r) \\
&- 3B^3(r)\beta^2(r)\frac{d}{dr}B(r) \\
&+ B(r)\frac{d}{dr}B(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
\hat{G}_{\hat{\varphi}\hat{\varphi}} = \frac{1}{rB^4(r)}\Big(&-rB^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r) \\
&- rB^4(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
&- 2rB^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\
&- 5rB^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
&- rB^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
&+ rB(r)\frac{d^2}{dr^2}B(r) \\
&- r\left(\frac{d}{dr}B(r)\right)^2 \\
&- 2B^4(r)\beta(r)\frac{d}{dr}\beta(r) \\
&- 3B^3(r)\beta^2(r)\frac{d}{dr}B(r) \\
&+ B(r)\frac{d}{dr}B(r)\Big)
\end{aligned}
$$

9 Energy Conditions

From $G_{\hat{a}\hat{b}} = 8\pi T_{\hat{a}\hat{b}}$ the stress-energy components in the orthonormal frame are:

9.1 Stress-Energy Components

Physical stress-energy

$$
\begin{aligned}
\rho = \frac{1}{8\pi r^2 B^4(r)} \Big(& 2r^2 B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
& + 3r^2 B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
& - 2r^2 B(r)\frac{d^2}{dr^2}B(r) \\
& + r^2\left(\frac{d}{dr}B(r)\right)^2 \\
& + 2rB^4(r)\beta(r)\frac{d}{dr}\beta(r) \\
& + 4rB^3(r)\beta^2(r)\frac{d}{dr}B(r) \\
& - 4rB(r)\frac{d}{dr}B(r) \\
& + B^4(r)\beta^2(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
p_1 = \frac{1}{8\pi r^2 B^4(r)} \Big(& -2r^2 B^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\
& - 2r^2 B^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
& - r^2 B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
& + r^2\left(\frac{d}{dr}B(r)\right)^2 \\
& - 2rB^4(r)\beta(r)\frac{d}{dr}\beta(r) \\
& - 6rB^3(r)\beta^2(r)\frac{d}{dr}B(r) \\
& + 2rB(r)\frac{d}{dr}B(r) \\
& - B^4(r)\beta^2(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
p_2 = \frac{1}{8\pi r B^4(r)} \Big(& -rB^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r) \\
& - rB^4(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
& - 2rB^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\
& - 5rB^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
& - rB^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
& + rB(r)\frac{d^2}{dr^2}B(r)
\end{aligned}
$$

$$- r\left(\frac{d}{dr}B(r)\right)^2$$
$$- 2B^4(r)\beta(r)\frac{d}{dr}\beta(r)$$
$$- 3B^3(r)\beta^2(r)\frac{d}{dr}B(r)$$
$$+ B(r)\frac{d}{dr}B(r)\Big)$$

$$p_3 = \frac{1}{8\pi r B^4(r)}\Big(-rB^4(r)\beta(r)\frac{d^2}{dr^2}\beta(r)$$
$$- rB^4(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$- 2rB^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r)$$
$$- 5rB^3(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$- rB^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$+ rB(r)\frac{d^2}{dr^2}B(r)$$
$$- r\left(\frac{d}{dr}B(r)\right)^2$$
$$- 2B^4(r)\beta(r)\frac{d}{dr}\beta(r)$$
$$- 3B^3(r)\beta^2(r)\frac{d}{dr}B(r)$$
$$+ B(r)\frac{d}{dr}B(r)\Big)$$

9.2 Null Energy Condition (NEC)

NEC requires $\rho + p_i \geq 0$ for all principal pressures p_i. Violation implies exotic (negative-energy) matter.

$$\rho + p_1 = \frac{1}{4\pi r B^4(r)}\Big(-rB^3(r)\beta^2(r)\frac{d^2}{dr^2}B(r)$$
$$+ rB^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$- rB(r)\frac{d^2}{dr^2}B(r)$$
$$+ r\left(\frac{d}{dr}B(r)\right)^2$$
$$- B^3(r)\beta^2(r)\frac{d}{dr}B(r)$$
$$- B(r)\frac{d}{dr}B(r)\Big)$$

$$\rho + p_2 = \frac{1}{8\pi r^2 B^3(r)}\Big(-r^2B^3(r)\beta(r)\frac{d^2}{dr^2}\beta(r)$$

$$
\begin{aligned}
&- r^2 B^3(r) \left(\frac{d}{dr}\beta(r)\right)^2 \\
&- 2r^2 B^2(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\
&- 3r^2 B^2(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
&+ 2r^2 B(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
&- r^2\frac{d^2}{dr^2}B(r) \\
&+ rB^2(r)\beta^2(r)\frac{d}{dr}B(r) \\
&- 3r\frac{d}{dr}B(r) \\
&+ B^3(r)\beta^2(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
\rho + p_3 = \frac{1}{8\pi r^2 B^3(r)}\Big(&-r^2 B^3(r)\beta(r)\frac{d^2}{dr^2}\beta(r) \\
&- r^2 B^3(r) \left(\frac{d}{dr}\beta(r)\right)^2 \\
&- 2r^2 B^2(r)\beta^2(r)\frac{d^2}{dr^2}B(r) \\
&- 3r^2 B^2(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r) \\
&+ 2r^2 B(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2 \\
&- r^2\frac{d^2}{dr^2}B(r) \\
&+ rB^2(r)\beta^2(r)\frac{d}{dr}B(r) \\
&- 3r\frac{d}{dr}B(r) \\
&+ B^3(r)\beta^2(r)\Big)
\end{aligned}
$$

9.3 Weak Energy Condition (WEC)

WEC: $\rho \geq 0$ (energy density non-negative) and NEC.

9.4 Strong Energy Condition (SEC)

SEC: $\rho + \sum_i p_i \geq 0$.

$$
\begin{aligned}
\rho + \textstyle\sum_i p_i = \frac{1}{4\pi r B(r)}\Big(&-rB(r)\beta(r)\frac{d^2}{dr^2}\beta(r) \\
&- rB(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
&- 3r\beta^2(r)\frac{d^2}{dr^2}B(r) \\
&- 5r\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)
\end{aligned}
$$

$$
\begin{aligned}
&- 2B(r)\beta(r)\frac{d}{dr}\beta(r) \\
&- 4\beta^2(r)\frac{d}{dr}B(r)\Big)
\end{aligned}
$$

9.5 Dominant Energy Condition (DEC)

DEC: $\rho \geq |p_i|$ — energy flows causally, no superluminal energy transport. *(The DEC requires knowledge of the sign of ρ relative to $|p_i|$; substitute explicit parameter values to evaluate.)*

10 Geodesic Equations

The geodesic equation for an affinely parameterised curve $x^\mu(\lambda)$ is

$$
\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu{}_{\alpha\beta}\,\dot{x}^\alpha\,\dot{x}^\beta = 0, \qquad \dot{x}^\mu \equiv \frac{dx^\mu}{d\lambda}. \tag{8}
$$

Each component written as $\ddot{x}^\mu = \ldots$:

$$
\begin{aligned}
\ddot{t} = -(& \\
&- \dot{\theta}^2 r^2 B(r)\beta(r)\frac{d}{dr}B(r) \\
&- \dot{\theta}^2 r B^2(r)\beta(r) \\
&- \dot{\varphi}^2 r^2 B(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r) \\
&- \dot{\varphi}^2 r B^2(r)\beta(r)\sin^2(\theta) \\
&- \dot{r}^2 B^2(r)\frac{d}{dr}\beta(r) \\
&- \dot{r}^2 B(r)\beta(r)\frac{d}{dr}B(r) \\
&- 2\dot{r}\dot{t}B^2(r)\beta(r)\frac{d}{dr}\beta(r) \\
&- 2\dot{r}\dot{t}B(r)\beta^2(r)\frac{d}{dr}B(r) \\
&- \dot{t}^2 B^2(r)\beta^2(r)\frac{d}{dr}\beta(r) \\
&- \dot{t}^2 B(r)\beta^3(r)\frac{d}{dr}B(r))
\end{aligned}
$$

$$
\ddot{r} = -(\frac{\dot{\theta}^2 r^2 B^2(r)\beta^2(r)\frac{d}{dr}B(r) - \dot{\theta}^2 r^2\frac{d}{dr}B(r) + \dot{\theta}^2 r B^3(r)\beta^2(r) - \dot{\theta}^2 r B(r) + \dot{\varphi}^2 r^2 B^2(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r) - \dot{\varphi}^2 r^2
$$

$$
\ddot{\theta} = -(\frac{2\dot{\theta}\dot{r}r\frac{d}{dr}B(r) + 2\dot{\theta}\dot{r}B(r) - \dot{\varphi}^2 r B(r)\sin(\theta)\cos(\theta)}{rB(r)})
$$

$$
\ddot{\varphi} = -(\frac{2\dot{\theta}\dot{\varphi}rB(r)\cos(\theta) + 2\dot{\varphi}\dot{r}r\sin(\theta)\frac{d}{dr}B(r) + 2\dot{\varphi}\dot{r}B(r)\sin(\theta)}{rB(r)\sin(\theta)})
$$

10.1 Conserved Quantities (Killing Vectors)

Coordinates that do not appear in $g_{\mu\nu}$ give rise to Killing vectors and conserved momenta along geodesics:

- $\partial/\partial t$ is a Killing vector. Conserved quantity: $p_t = g_{t\nu}\,\dot{x}^\nu = \text{const.}$
- $\partial/\partial\varphi$ is a Killing vector. Conserved quantity: $p_\varphi = g_{\varphi\nu}\,\dot{x}^\nu = \text{const.}$

11 Conservation and Consistency Checks

11.1 Contracted Bianchi Identity: $\nabla_\mu G^{\mu\nu} = 0$

This is a geometric identity that guarantees stress-energy conservation $\nabla_\mu T^{\mu\nu} = 0$ via the field equations.

Bianchi Identity — Simplification Incomplete

Note: SymPy could not reduce all components of $\nabla_\mu G^{\mu\nu}$ to zero using `cancel()` followed by `simplify()`. This is *not necessarily a code error* — it is a known limitation of SymPy for metrics containing unspecified symbolic functions such as $B(r)$, $\beta(r)$, $f(r)$, etc.

What this means physically: The Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ is guaranteed to hold for *any* smooth metric by the second Bianchi identity for the Riemann tensor — it is a mathematical identity, not an equation of motion. If the residuals below are large symbolic expressions involving derivatives of abstract metric functions, they almost certainly vanish but SymPy lacks a simplification rule to confirm it automatically.

What to do: (1) Substitute a *specific* form of every metric function (e.g. set $B(r) = 1$ and $\beta(r) = \sqrt{2M/r}$ for Schwarzschild-PG) — the residuals will collapse to zero numerically. (2) If the residuals contain only small *rational* numbers (not long symbolic sums), that would indicate a genuine computation error and should be reported.

$$
\begin{aligned}
\nabla_\mu G^{\mu t} = \frac{1}{r^2 B^3(r)} \Big(& r^4 B^4(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d^2}{dr^2}\beta(r) \\
& + r^4 B^4(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}\beta(r) \\
& + r^4 B^4(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
& + r^4 B^4(r)\beta(r)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
& + 2r^4 B^3(r)\beta^3(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r) \\
& + 2r^4 B^3(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r) \\
& + 5r^4 B^3(r)\beta^2(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r) \\
& + 5r^4 B^3(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r) \\
& + r^4 B^2(r)\beta^3(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^3
\end{aligned}
$$

$$+ r^4 B^2(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^3$$
$$- r^4 B(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$
$$- r^4 B(r)\beta(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$
$$+ r^4\beta(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^3$$
$$+ r^4\beta(r)\left(\frac{d}{dr}B(r)\right)^3$$
$$+ r^3 B^5(r)\beta^2(r)\sin^2(\theta)\frac{d^2}{dr^2}\beta(r)$$
$$+ r^3 B^5(r)\beta^2(r)\frac{d^2}{dr^2}\beta(r)$$
$$+ r^3 B^5(r)\beta(r)\sin^2(\theta)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ r^3 B^5(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ 2r^3 B^4(r)\beta^3(r)\sin^2(\theta)\frac{d^2}{dr^2}B(r)$$
$$+ 2r^3 B^4(r)\beta^3(r)\frac{d^2}{dr^2}B(r)$$
$$+ 7r^3 B^4(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$+ 7r^3 B^4(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$+ 4r^3 B^3(r)\beta^3(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2$$
$$+ 4r^3 B^3(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$- r^3 B^2(r)\beta(r)\sin^2(\theta)\frac{d^2}{dr^2}B(r)$$
$$- r^3 B^2(r)\beta(r)\frac{d^2}{dr^2}B(r)$$
$$+ 2r^2 B^5(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}\beta(r)$$
$$+ 2r^2 B^5(r)\beta^2(r)\frac{d}{dr}\beta(r)$$
$$+ 2r^2 B^4(r)\beta^4(r)\frac{d^2}{dr^2}B(r)\frac{d}{dr}\beta(r)$$
$$+ 3r^2 B^4(r)\beta^3(r)\sin^2(\theta)\frac{d}{dr}B(r)$$
$$+ 2r^2 B^4(r)\beta^3(r)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ 3r^2 B^4(r)\beta^3(r)\frac{d}{dr}B(r)$$
$$+ 2r^2 B^4(r)\beta^2(r)\frac{d^2}{dr^2}B(r)\frac{d}{dr}\beta(r)$$
$$+ 2r^2 B^4(r)\beta(r)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2$$

$$
\begin{aligned}
&+ 2r^2B^3(r)\beta^5(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)\\
&+ 3r^2B^3(r)\beta^4(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)\\
&+ 2r^2B^3(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)\\
&+ 3r^2B^3(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)\\
&+ r^2B^2(r)\beta^5(r)\left(\frac{d}{dr}B(r)\right)^3\\
&+ r^2B^2(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^3\\
&- 2r^2B^2(r)\beta^2(r)\frac{d^2}{dr^2}B(r)\frac{d}{dr}\beta(r)\\
&- r^2B^2(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)\\
&- r^2B^2(r)\beta(r)\frac{d}{dr}B(r)\\
&- 2r^2B(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)\\
&+ r^2B(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)\\
&- r^2B(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)\\
&+ r^2\beta^3(r)\left(\frac{d}{dr}B(r)\right)^3\\
&- r^2\beta(r)\left(\frac{d}{dr}B(r)\right)^3\\
&+ 2rB^5(r)\beta^3(r)\left(\frac{d}{dr}\beta(r)\right)^2\\
&+ 2rB^5(r)\beta(r)\left(\frac{d}{dr}\beta(r)\right)^2\\
&+ 8rB^4(r)\beta^4(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&+ 8rB^4(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&+ 6rB^3(r)\beta^5(r)\left(\frac{d}{dr}B(r)\right)^2\\
&+ 6rB^3(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2\\
&- 4rB^2(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&- 2rB^2(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&- 4rB(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2\\
&- 2rB(r)\beta(r)\left(\frac{d}{dr}B(r)\right)^2
\end{aligned}
$$

$$+ B^5(r)\beta^4(r)\frac{d}{dr}\beta(r)$$
$$+ B^5(r)\beta^2(r)\frac{d}{dr}\beta(r)$$
$$+ B^4(r)\beta^5(r)\frac{d}{dr}B(r)$$
$$+ B^4(r)\beta^3(r)\frac{d}{dr}B(r)\Big)$$

$$\nabla_\mu G^{\mu r} = \frac{1}{r^2 B^5(r)}\Big(-r^4 B^6(r)\beta^3(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d^2}{dr^2}\beta(r)$$
$$- r^4 B^6(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}\beta(r)$$
$$- r^4 B^6(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$- r^4 B^6(r)\beta^2(r)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$- 2r^4 B^5(r)\beta^4(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$
$$- 2r^4 B^5(r)\beta^4(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$
$$- 5r^4 B^5(r)\beta^3(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)$$
$$- 5r^4 B^5(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)$$
$$- r^4 B^4(r)\beta^4(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^3$$
$$- r^4 B^4(r)\beta^4(r)\left(\frac{d}{dr}B(r)\right)^3$$
$$+ r^4 B^4(r)\beta(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d^2}{dr^2}\beta(r)$$
$$+ r^4 B^4(r)\beta(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}\beta(r)$$
$$+ r^4 B^4(r)\sin^2(\theta)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ r^4 B^4(r)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ 3r^4 B^3(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$
$$+ 3r^4 B^3(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$
$$+ 5r^4 B^3(r)\beta(r)\sin^2(\theta)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)$$
$$+ 5r^4 B^3(r)\beta(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)$$
$$- r^4 B(r)\sin^2(\theta)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$
$$- r^4 B(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$

$$+ r^4 \sin^2(\theta) \left(\frac{d}{dr}B(r)\right)^3$$
$$+ r^4 \left(\frac{d}{dr}B(r)\right)^3$$
$$- r^3 B^7(r) \beta^3(r) \sin^2(\theta) \frac{d^2}{dr^2}\beta(r)$$
$$- r^3 B^7(r) \beta^3(r) \frac{d^2}{dr^2}\beta(r)$$
$$- r^3 B^7(r) \beta^2(r) \sin^2(\theta) \left(\frac{d}{dr}\beta(r)\right)^2$$
$$- r^3 B^7(r) \beta^2(r) \left(\frac{d}{dr}\beta(r)\right)^2$$
$$- 2r^3 B^6(r) \beta^4(r) \sin^2(\theta) \frac{d^2}{dr^2}B(r)$$
$$- 2r^3 B^6(r) \beta^4(r) \frac{d^2}{dr^2}B(r)$$
$$- 7r^3 B^6(r) \beta^3(r) \sin^2(\theta) \frac{d}{dr}B(r) \frac{d}{dr}\beta(r)$$
$$- 7r^3 B^6(r) \beta^3(r) \frac{d}{dr}B(r) \frac{d}{dr}\beta(r)$$
$$- 4r^3 B^5(r) \beta^4(r) \sin^2(\theta) \left(\frac{d}{dr}B(r)\right)^2$$
$$- 4r^3 B^5(r) \beta^4(r) \left(\frac{d}{dr}B(r)\right)^2$$
$$+ r^3 B^5(r) \beta(r) \sin^2(\theta) \frac{d^2}{dr^2}\beta(r)$$
$$+ r^3 B^5(r) \beta(r) \frac{d^2}{dr^2}\beta(r)$$
$$+ r^3 B^5(r) \sin^2(\theta) \left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ r^3 B^5(r) \left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ 3r^3 B^4(r) \beta^2(r) \sin^2(\theta) \frac{d^2}{dr^2}B(r)$$
$$+ 3r^3 B^4(r) \beta^2(r) \frac{d^2}{dr^2}B(r)$$
$$+ 7r^3 B^4(r) \beta(r) \sin^2(\theta) \frac{d}{dr}B(r) \frac{d}{dr}\beta(r)$$
$$+ 7r^3 B^4(r) \beta(r) \frac{d}{dr}B(r) \frac{d}{dr}\beta(r)$$
$$+ 4r^3 B^3(r) \beta^2(r) \sin^2(\theta) \left(\frac{d}{dr}B(r)\right)^2$$
$$+ 4r^3 B^3(r) \beta^2(r) \left(\frac{d}{dr}B(r)\right)^2$$
$$- r^3 B^2(r) \sin^2(\theta) \frac{d^2}{dr^2}B(r)$$
$$- r^3 B^2(r) \frac{d^2}{dr^2}B(r)$$

$$
\begin{aligned}
& - 2r^2 B^7(r)\beta^3(r)\sin^2(\theta)\frac{d}{dr}\beta(r) \\
& - 2r^2 B^7(r)\beta^3(r)\frac{d}{dr}\beta(r) \\
& - 2r^2 B^6(r)\beta^5(r)\frac{d^2}{dr^2}B(r)\frac{d}{dr}\beta(r) \\
& - 3r^2 B^6(r)\beta^4(r)\sin^2(\theta)\frac{d}{dr}B(r) \\
& - 2r^2 B^6(r)\beta^4(r)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
& - 3r^2 B^6(r)\beta^4(r)\frac{d}{dr}B(r) \\
& - 2r^2 B^6(r)\beta^3(r)\frac{d^2}{dr^2}B(r)\frac{d}{dr}\beta(r) \\
& - 2r^2 B^6(r)\beta^2(r)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
& - 2r^2 B^5(r)\beta^6(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r) \\
& - 3r^2 B^5(r)\beta^5(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r) \\
& - 2r^2 B^5(r)\beta^4(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r) \\
& - 3r^2 B^5(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r) \\
& + 2r^2 B^5(r)\beta(r)\sin^2(\theta)\frac{d}{dr}\beta(r) \\
& + 2r^2 B^5(r)\beta(r)\frac{d}{dr}\beta(r) \\
& - r^2 B^4(r)\beta^6(r)\left(\frac{d}{dr}B(r)\right)^3 \\
& - r^2 B^4(r)\beta^4(r)\left(\frac{d}{dr}B(r)\right)^3 \\
& + 4r^2 B^4(r)\beta^3(r)\frac{d^2}{dr^2}B(r)\frac{d}{dr}\beta(r) \\
& + 4r^2 B^4(r)\beta^2(r)\sin^2(\theta)\frac{d}{dr}B(r) \\
& + 2r^2 B^4(r)\beta^2(r)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
& + 4r^2 B^4(r)\beta^2(r)\frac{d}{dr}B(r) \\
& - 2r^2 B^4(r)\beta(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}\beta(r) \\
& - 2r^2 B^4(r)\beta(r)\frac{d^2}{dr^2}B(r)\frac{d}{dr}\beta(r) \\
& - 2r^2 B^4(r)\frac{d}{dr}B(r)\left(\frac{d}{dr}\beta(r)\right)^2 \\
& + 4r^2 B^3(r)\beta^4(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r) \\
& + 2r^2 B^3(r)\beta^3(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)
\end{aligned}
$$

$$- 6r^2B^3(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$
$$- 11r^2B^3(r)\beta(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)$$
$$- 5r^2B^2(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^3$$
$$- 2r^2B^2(r)\beta(r)\frac{d^2}{dr^2}B(r)\frac{d}{dr}\beta(r)$$
$$- r^2B^2(r)\sin^2(\theta)\frac{d}{dr}B(r)$$
$$- r^2B^2(r)\frac{d}{dr}B(r)$$
$$- 2r^2B(r)\beta^2(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$
$$+ r^2B(r)\beta(r)\left(\frac{d}{dr}B(r)\right)^2\frac{d}{dr}\beta(r)$$
$$+ 2r^2B(r)\frac{d}{dr}B(r)\frac{d^2}{dr^2}B(r)$$
$$+ r^2\beta^2(r)\left(\frac{d}{dr}B(r)\right)^3$$
$$- 2rB^7(r)\beta^4(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$- 2rB^7(r)\beta^2(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$- 8rB^6(r)\beta^5(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$- 8rB^6(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$- 6rB^5(r)\beta^6(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$- 6rB^5(r)\beta^4(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$+ 2rB^5(r)\beta^2(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$- 2rB^5(r)\beta(r)\frac{d^2}{dr^2}\beta(r)$$
$$- 2rB^5(r)\left(\frac{d}{dr}\beta(r)\right)^2$$
$$+ 12rB^4(r)\beta^3(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$- 4rB^4(r)\beta^2(r)\frac{d^2}{dr^2}B(r)$$
$$- 18rB^4(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)$$
$$+ 10rB^3(r)\beta^4(r)\left(\frac{d}{dr}B(r)\right)^2$$
$$- 16rB^3(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2$$

$$
\begin{aligned}
&- 4rB^2(r)\beta(r)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&+ 2rB^2(r)\frac{d^2}{dr^2}B(r)\\
&- 4rB(r)\beta^2(r)\left(\frac{d}{dr}B(r)\right)^2\\
&+ 4rB(r)\left(\frac{d}{dr}B(r)\right)^2\\
&- B^7(r)\beta^5(r)\frac{d}{dr}\beta(r)\\
&- B^7(r)\beta^3(r)\frac{d}{dr}\beta(r)\\
&- B^6(r)\beta^6(r)\frac{d}{dr}B(r)\\
&- B^6(r)\beta^4(r)\frac{d}{dr}B(r)\\
&+ B^5(r)\beta^3(r)\frac{d}{dr}\beta(r)\\
&- 4B^5(r)\beta(r)\frac{d}{dr}\beta(r)\\
&+ B^4(r)\beta^4(r)\frac{d}{dr}B(r)\\
&- 8B^4(r)\beta^2(r)\frac{d}{dr}B(r)\\
&+ 2B^2(r)\frac{d}{dr}B(r)\Big)
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu G^{\mu\theta} = {}& \frac{1}{rB^4(r)\sin(\theta)}\Big(-rB^4(r)\beta(r)\cos^3(\theta)\frac{d^2}{dr^2}\beta(r)\\
&- rB^4(r)\cos^3(\theta)\left(\frac{d}{dr}\beta(r)\right)^2\\
&- 2rB^3(r)\beta^2(r)\cos^3(\theta)\frac{d^2}{dr^2}B(r)\\
&- 5rB^3(r)\beta(r)\cos^3(\theta)\frac{d}{dr}B(r)\frac{d}{dr}\beta(r)\\
&- rB^2(r)\beta^2(r)\cos^3(\theta)\left(\frac{d}{dr}B(r)\right)^2\\
&+ rB(r)\cos^3(\theta)\frac{d^2}{dr^2}B(r)\\
&- r\cos^3(\theta)\left(\frac{d}{dr}B(r)\right)^2\\
&- 2B^4(r)\beta(r)\cos^3(\theta)\frac{d}{dr}\beta(r)\\
&- 3B^3(r)\beta^2(r)\cos^3(\theta)\frac{d}{dr}B(r)\\
&+ B(r)\cos^3(\theta)\frac{d}{dr}B(r)\Big)
\end{aligned}
$$

11.2 Ricci Scalar Trace Consistency

Verify $g^{\mu\nu}R_{\mu\nu} = \mathcal{R}$ (cross-check).

Trace Check

VERIFIED: $g^{\mu\nu} R_{\mu\nu} = \mathcal{R}$. ✓

11.3 Metric Inverse Identity

Verify $g^{\mu\rho} \, g_{\rho\nu} = \delta^{\mu}{}_{\nu}$.

Metric Inverse

VERIFIED: $g^{\mu\rho} \, g_{\rho\nu} = \delta^{\mu}{}_{\nu}$. ✓